

Autonomous Driving Cars in Smart Cities: Recent Advances, Requirements, and Challenges

Ibrar Yaqoob, Latif U. Khan, S. M. Ahsan Kazmi, Muhammad Imran, Nadra Guizani, and Choong Seon Hong

ABSTRACT

An unprecedented proliferation of autonomous driving technologies has been observed in recent years, resulting in the emergence of reliable and safe transportation services. In the foreseeable future, millions of autonomous cars will communicate with each other and become prevalent in smart cities. Thus, scalable, robust, secure, fault-tolerant, and interoperable technologies are required to support such a plethora of autonomous cars. In this article, we investigate, highlight, and report premier research advances made in autonomous driving by devising a taxonomy. A few indispensable requirements for successful deployment of autonomous cars are enumerated and discussed. Furthermore, we discover and present recent synergies and prominent case studies on autonomous driving. Finally, several imperative open research challenges are identified and discussed as future research directions.

INTRODUCTION

The recent proliferation of miniaturized autonomous driving technologies has revolutionized cities by making smart cars a viable option for daily transportation. Autonomous cars alleviate human drivers' burden by performing intelligent operations, such as collision avoidance, lane departure warning, and traffic sign detection. In addition, autonomous driving technologies can efficiently manage traffic flow and reduce congestion and have advanced fuel economy by lowering emissions [1]. Autonomous cars assist people in their daily lives by providing reliable and safe transportation services to elderly and disabled people, handling parking problems, and eliminating a substantial number of accidents previously caused by human errors [2]. Figure 1 illustrates the concept of autonomous driving cars in smart cities.

Gartner predicts that 250 million cars will be connected with each other by the end of 2020. Another report reveals that the artificial intelligence (AI) market is expected to be valued at \$11,000 million by 2025 (accessed on: 25 Oct. 2018 <https://medium.com/datadriveninvestor/artificial-intelligence-and-autonomous-vehicles-ae877feb6cd2>). IHS Markit anticipates that the installation rate of AI-based systems will grow up to 109 percent in 2025. McKinsey estimates that autonomous cars will produce a substantial reve-

nue reaching \$450 billion to \$750 billion by 2030 (accessed on: 28 October 2018 <https://itpeernet-work.intel.com/5g-key-fully-realizing-connected-autonomous-vehicles/>). However, the development of autonomous cars requires contemporary solutions in terms of perception, planning, and control. Although autonomous cars are usually equipped with powerful computing and sensing technologies based on heterogeneous architectures, many inherent challenges associated with communication and networking technologies, privacy and security, real-time data analytics, data transmission, and limited bandwidth hinder autonomous cars from becoming a mainstream technology [3, 4]. Numerous research efforts have been performed to overcome the aforementioned challenges. In this study, we aim to explore and investigate the recent solutions and advances made in autonomous driving technology. The contributions are as follows.

- We investigate, highlight, and report premier research advances made in autonomous cars by devising a taxonomy.
- We enumerate and discuss indispensable requirements for the successful deployment of autonomous cars.
- We discover and present recent synergies and prominent case studies on autonomous driving.
- We identify and discuss future research challenges.

These contributions are provided in separate sections, and the concluding remarks are provided in the final section.

RECENT ADVANCES

Figure 2 elucidates premier research advances in the form of a taxonomy, which is devised on the basis of essential parameters, such as levels of autonomous driving, communication technologies, components, objectives, required capabilities, artificial intelligence, and emerging technologies.

LEVELS OF AUTONOMOUS DRIVING

Automated driving can be defined using five levels [5]. In level 1, most components are controlled by humans, but certain functions, such as steering and accelerating, can be automatically performed. In level 2, at least one driver assistance system is automated. However, the driver still needs to be ready at any time to take control of the vehicle. In level 3,

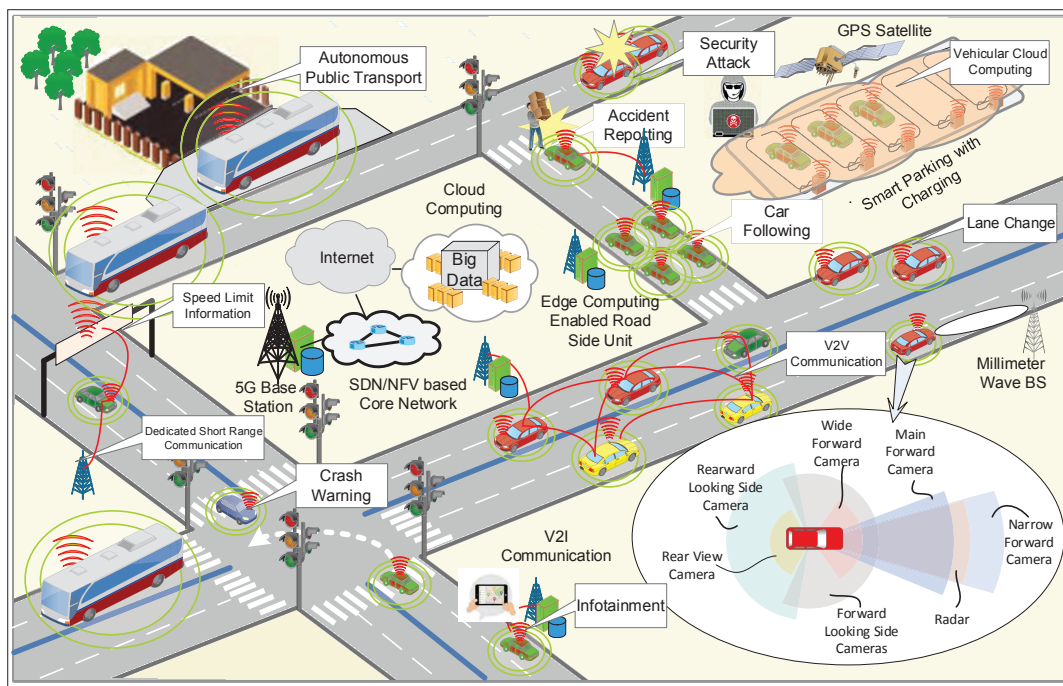


FIGURE 1. An illustration of autonomous driving cars in smart cities.

a driver is still needed in the car and can only intervene if necessary. Although level 4 is meant for fully autonomous cars, it only covers a few driving scenarios. Only level 5 refers to fully autonomous cars expected to show a performance equal to that of a human driver in every driving scenario.

COMMUNICATION TECHNOLOGIES

The advent of modern communication technologies has transformed traditional transportation industries into smart transportation. Traditionally, vehicles exhibited limited ability to communicate with each other. However, the vision of autonomous cars is becoming a reality because of the fifth-generation (5G) standard, which ensures low latency and provides high data rates [6]. Autonomous cars send and receive data from multiple sources, such as traffic signals and parking spaces, autonomous vehicles, and the cloud, which demand highly reliable communication technologies. The key technologies involved in 5G are named millimeter-wave, beamforming, device-to-device communication, and small-cell technology. Autonomous cars have recently been researched in the context of dedicated short range communications (DSRC) and the cellular vehicle-to-everything standard. Several indispensable characteristics of these technologies, such as low latency, high throughput, network slicing, flexibility, high capacity, highly directional, shorter wavelengths, and affordability, make them remarkably favorable for autonomous cars [7]. Furthermore, the combination of 5G networks with network function virtualization (NFV) and software defined networking (SDN) makes it highly powerful for autonomous cars.

Long term evolution for vehicle (LTE-V) is a base station driven technology. LTE-V is currently based on 4G technology. However, LTE-V can be extended toward 5G in the future. LTE-V provides wide coverage and enables support for high-speed autonomous driving scenarios compared with DSRC. Wireless fidelity (WiFi) and DSRC belong to the IEEE 802.11 family. In autonomous cars, WiFi can

be used for information dissemination and Internet access [7]. DSRC is an upgraded technology of WiFi, which is highly efficient and specially designed for automotive applications. Bluetooth is a low-power, short-range, and low-cost communication technology. However, this technology is unsuitable for real-time autonomous vehicular applications.

The distinctive ZigBee features, such as low cost of deployment, easy maintenance, and long battery life, make it suitable for vehicle to vehicle and vehicle to infrastructure communication. Ultra wideband (UWB) is mostly used for intra-vehicle wireless interconnection [7]. Cooperative relaying and opportunistic routing between cars can help in overcoming intermittent communication infrastructure [8].

COMPONENTS

Seven integral components of autonomous cars include ultrasonic sensors, cameras, radars, infrared, lidars, local data processors, and global positioning system (GPS) devices. The ultrasonic sensor helps in detecting objects and determining distances. Onboard cameras ensure that the cars run in a safe manner. Radar uses radio wave technology to determine the distance, angle, and velocity of the surrounding objects. Infrared helps to provide night vision to self-driving cars. Lidar helps in measuring ranges and sensing brake lights. Local data processors enable automatic real-time calculations, thereby resulting in high-speed decision-making while traveling. GPS helps to track the car's location and guides it toward a set destination.

OBJECTIVES

Autonomous cars aim to reduce fuel consumption, accidents, and congestion; save time and space; and improve mobility for disabled and elderly people. In urban areas, people usually spend billion of hours in traffic, thereby resulting in wasted fuel. The technology involved in autonomous cars is developed in such a way that

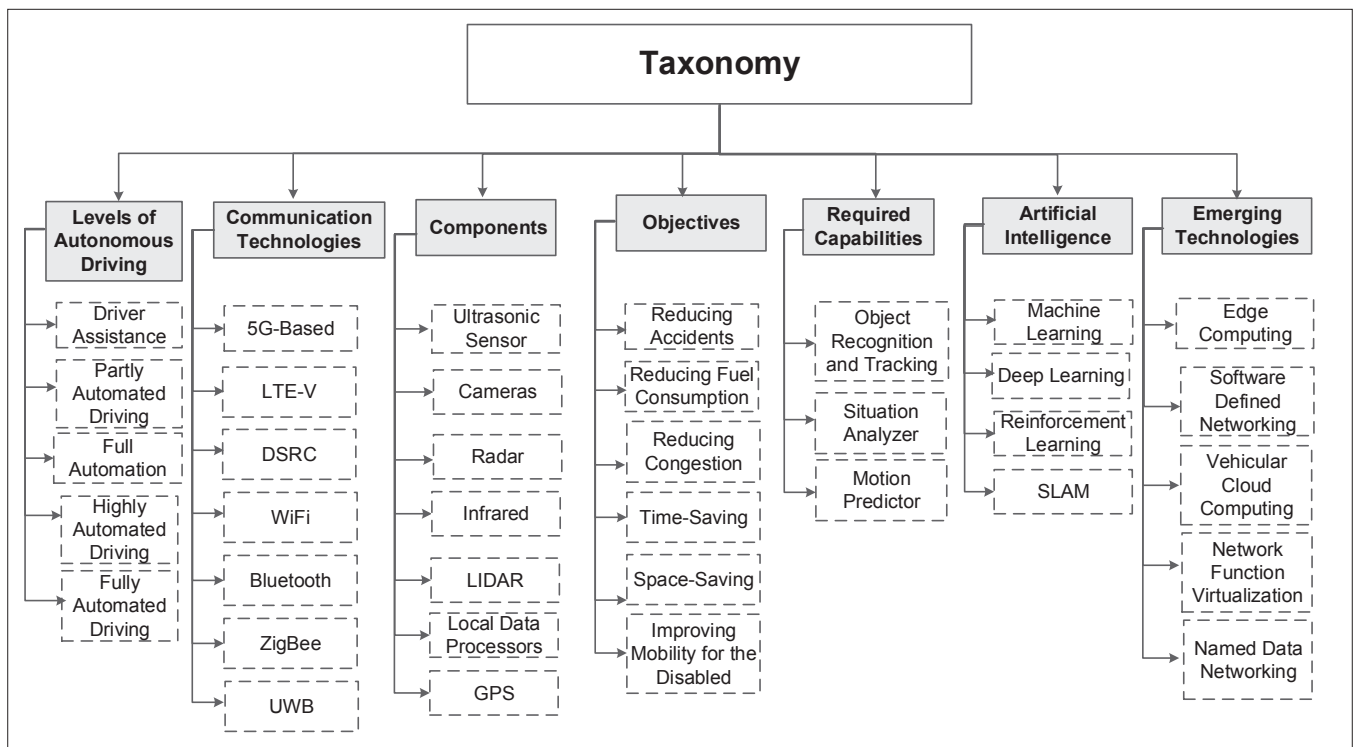


FIGURE 2. Taxonomy of autonomous driving cars.

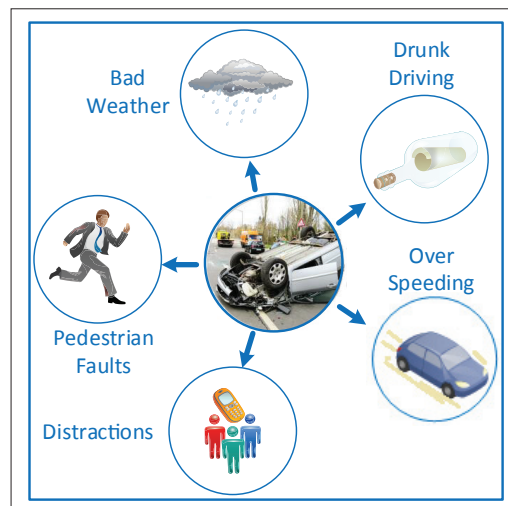


FIGURE 3. Fundamental causes of car accidents.

it can reduce fuel consumption because they are expected to reduce emissions by 60 percent. Autonomous cars can help in reducing a significant number of accidents which usually happen due to drunk driving, bad weather, pedestrian faults, over speeding, and human errors, as illustrated in Fig. 3. The successful deployment of an autonomous car will likely lead to 90 percent prevention of road accidents. Autonomous cars are likely to save up to an hour on a daily basis because human intervention is uninvolved. These cars are greatly capable of accessing up-to-the-minute data to help monitor traffic data and maps accordingly, thereby resulting in the determination of the highly efficient route. In this manner, severe congestion problems can be mitigated. Another objective of autonomous cars is to handle parking problems. Autonomous cars can be parked with

only a small space requirement, thereby resulting in significant space saving for urban communities. Improving mobility for disabled people is also an objective of autonomous cars.

REQUIRED CAPABILITIES

Although autonomous cars require numerous capabilities, three of them are highly critical. For example, an autonomous car must be able to effectively identify and track various objects, such as road signs, traffic lights, and lane markers, on the road that humans gain through their perception. A situation analyzer enables the cars to compute and analyze the voluminous amount of data in a real-time manner, which will help the motion predictor in making some intelligent decisions. A motion predictor is based on AI algorithms that help in handling critical situations that arise during driving, such as people running across roads and unpredictable weather conditions [9].

ARTIFICIAL INTELLIGENCE

AI techniques enable autonomous cars to manage and derive sense from the humongous amount of data produced by the cars (cameras, sensors, radar, lidar, communication systems, GPS, internal systems, and mechanical sensors) [10]. Autonomous cars usually coordinate with each other by sharing some data, such as location information and internal system status. Applying AI solutions on such data can enable the vehicles to see, hear, and make decisions, as shown in Fig. 4. Specifically, machine learning algorithms enable autonomous cars to monitor their surrounding environments and make predictions accordingly. These algorithms can be divided into four categories, such as regression algorithms, pattern recognition, cluster algorithms, and decision matrix algorithms. With regard to autonomous cars, deep learning technology can provide facilitation in terms of voice recognition, voice search, recommen-

dation engine, sentiment analysis, image recognition, and motion detection. Reinforcement learning helps in controlling motion and provides support in decision-making. Simultaneous localization and mapping (SLAM) is a category of techniques that an autonomous vehicle can use to map its area and determine its own location on a map [11]. In summary, AI-based solutions can enable the proper and safe operation of autonomous cars.

EMERGING TECHNOLOGIES

The emerging technologies for autonomous cars are as follows: edge computing, vehicular cloud computing (VCC), SDN, NFV, and named data networking (NDN). The rise of edge computing has enabled autonomous cars to efficiently process data and find patterns in a real-time manner by moving sensors' data close to the cars' network, thereby resulting in making rapid decision-making [12]. A recent study revealed that autonomous vehicles generate 0.75 GB of data in each second, which demands fast storage and transfer of data that can be coped with using edge computing.

SDN-based approaches can enable interoperability between heterogeneous data generated by electronic control modules of autonomous cars. Enabling interaction between these data sources can bring innovation and new smart features that can significantly improve security and riders' experience. The interoperability feature can enable autonomous cars to mitigate a high-risk situation using the SDN-based solutions. VCC brings a remarkable effect on autonomous cars in terms of traffic management and road safety by employing vehicular resources, namely, computing, storage, and Internet for decision-making purposes. NFV allows network functions to be distributed, focusing on computational power in such a way that they can be efficiently utilized. In terms of autonomous cars, NFV allows 5G-enabled vehicles to be focused on the services and locations that are mainly required. Moreover, NFV helps in enabling network slicing, thereby leading to the formation of multiple logical networks on top of the 5G-based vehicular network infrastructure where each slice is dedicated and secured for a specific function. One of the major advantages of network slicing is its ability to ensure that autonomous cars obtain access to critical data that need to be safely operated. NDN eliminates related issues posed by addresses and enables in-network caching and secure data sharing among autonomous cars [13]. Table 1 summarizes the benefits of these emerging technologies.

REQUIREMENTS

Figure 5 depicts the core requirements that need to be fulfilled for enabling autonomous driving cars. The rationale for the requirements and their possible solutions are presented in this section.

FAULT TOLERANCE

How can faults be identified, localized, and rectified in autonomous cars with particularly limited human intervention?

Autonomous cars use a collection of technologies, such as cameras, lidar, radar, and computer vision, along with software packages to deal with the control functions. However, the cars are exceedingly vulnerable to hardware or software failures because they incorporate modern technologies on

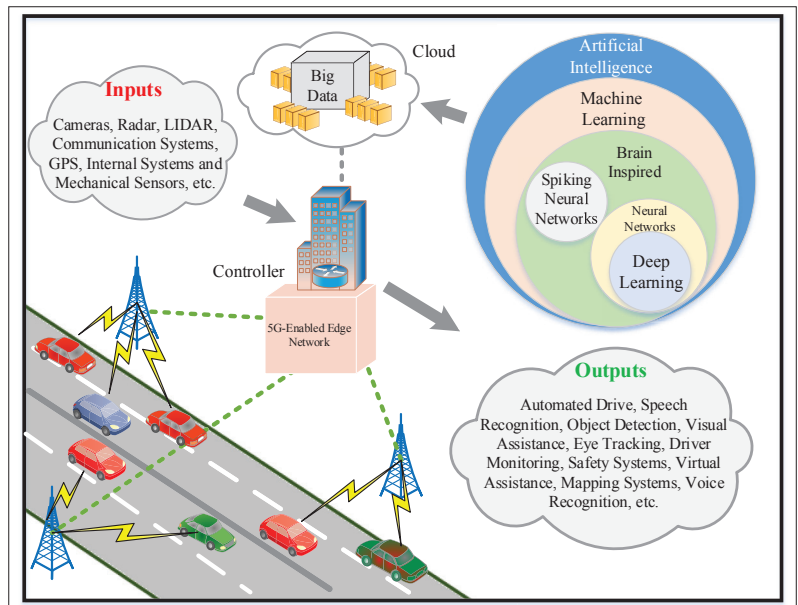


FIGURE 4. AI-enabled autonomous driving cars.

the basis of mechanical/electrical systems, which cause service outage for a certain amount of time and thus results in fatal accidents. The fault management process involves determining the fault location, restricting it, and subsequently removing it with a particularly limited human intervention. However, the implementation of the process engenders new challenges. Thus, novel techniques for fault localization followed by fault removal must be developed. Ensuring the use of robust sensors along with methodologies that incorporate the fault prevention measures is highly essential.

STRICT LATENCY

How can autonomous cars with strict latency-aware computation and storage capabilities be enabled to permit infotainment and big data analytics?

Edge computing pushes computation and storage resources to the network end for performing latency-sensitive tasks. Caching the content at roadside units tends to reduce delay instead of accessing the content in a remote manner. Edge computing along with caching can play a critical role in enabling infotainment and big data analytics in autonomous driving. Autonomous cars generate a massive amount of data. Most data are of an unstructured type, whose processing requires powerful analytics for generating actionable data. Infotainment services can also be efficiently provided in the cars using edge computing and caching, given that cloud computing is no longer a feasible option due to its remote location. However, designing and developing such algorithms that can efficiently provide infotainment services by employing edge computing and caching technologies within autonomous cars are highly challenging.

ROBUST ARCHITECTURE

How can a robust architecture that effectively controls heterogeneous actuators, main computers, and sensors with low complexity be designed?

Vehicles must perform the management of complex operations, including localization, plan-

Emerging technologies	Role in autonomous driving cars
Edge computing	• Offers storage and real-time data processing facilities. • Provides caching services which lead to offering infotainment services. • Non-critical data can be offloaded, and only critical data are kept within the vehicle.
Software defined networking	• Ensures optimal resource utilization. • Makes vehicular networks flexible and scalable. • Eliminates the static nature of traditional networks.
Vehicular cloud computing	• Efficient road traffic management. • Ensures road safety by instantly using vehicular resources. • On-demand elastic application services. • Dynamic traffic light management.
Network function virtualization	• Helps in enabling network slicing. • Provides efficiencies in terms of space, power, and cooling. • Allows network functions to be distributed.
Named data networking	• Eliminates issues posed by the addresses, such as address space exhaustion, network address translation traversal, mobility, and address management, in the IP architecture. • Naming data enables automatic in-network caching. • Provides an enhanced foundation for secure data sharing among autonomous cars.

TABLE 1. Role of emerging technologies in autonomous driving cars.

ning, control, and driving, along with heterogeneous sensors to enable autonomous driving. Generally, autonomous vehicles are equipped with mobile robot technology, autonomous industry, computing technologies (such as cloud computing and edge computing), and communication technologies. Each technology possesses its own specifications and requirements. Therefore, novel and robust architectures are required to enable and integrate these technologies in autonomous vehicles. The architecture of autonomous vehicles can be either centralized or decentralized. In a centralized system, the centralized entity controls sensors and actuators of autonomous vehicles. In a decentralized approach, sensors and actuators are implemented into several sub-units. The centralized system exhibits easy management but high computation complexity compared with the decentralized architecture. In autonomous driving, the scale of the overall system is expected to be large. Therefore, a decentralized approach is highly suitable to enable a system with low computation complexity and robustness. Apart from complexity, the decentralized nature further aids the fault tolerance of the system, which is one of the prime concerns in autonomous driving. Meanwhile, the architecture of autonomous vehicles along with other roadside units must be designed in such a way that it can enable efficient communication between autonomous vehicles, cloud, and edge nodes using access networks.

RESOURCE MANAGEMENT

How can the communication, computation, storage, and energy resources be efficiently managed among autonomous cars?

In autonomous driving, communication resources ensure connectivity between cars. These resources must be optimally utilized, which can be ensured using the different schemes based on optimization,

game theory, and hybrid approaches. Although edge computing offers computation and storage services to autonomous cars, it poses new challenges in terms of management and functionalities. One of the possible options with respect to energy resources is to efficiently employ potential and kinetic energy storage for attaining adaptive energy speed for enabling economic driving of autonomous vehicles. For example, reducing speed while climbing up a mountain and increasing it when driving downward are preferable. In the future, effective speed control schemes should be designed for enabling efficient operations of autonomous vehicles.

LOCALIZATION

How can the exact location of autonomous cars be precisely determined for path planning and decision making?

The localization process consists of two phases: environmental information collection using sensors and then precisely estimating their location. In a complex scenario, computing the location of a vehicle using internal measurement units and GPS is a challenging task and may result in inaccurate readings due to a lack of an adequate number of visible satellites. Localization schemes using lidar can also be used. However, the high cost associated with lidar sensors poses a limitation on its practical use. Therefore, new schemes that use low-cost hardware and generate precise results in dynamic urban environments must be developed.

SECURITY AND PRIVACY

How can autonomous cars be prevented from obtaining unauthorized access to avoid accidents?

Among the levels of car automation, level 5 is highly vulnerable to security threats because it is intended to enable driving without a driver. A hacker may hijack the car and use it to cause accidents and injuries. Apart from accidents, the hacker can lock the passengers inside the car to keep them for ransom money. Generally, potential/possible attacks on self-driving cars can be divided into two categories: physical access attacks and man-in-the-middle attacks. The hacker can physically access the car and install malicious software in it. In addition, the hacker can alter communication signals. Novel and effective authentication algorithms should be designed to avoid unauthorized access for enabling secure autonomous driving.

AUTONOMOUS DRIVING SYNERGIES AND CASE STUDIES

This section briefly describes the reported case studies of autonomous cars by different manufacturers. An overview of the features and different sensors used in four case studies is provided in this section.

MERCEDES-BENZ F 015 LUXURY IN MOTION

The Mercedes-Benz F 015 (accessed on: 23 October 2018 <https://media.daimler.com/mars-MediaSite/en/instance/ko/Overview-Mercedes-Benz-F015-Luxury-in-Motion.xhtml?oid=9904624>) is an autonomous research vehicle whose idea was conceived by the team from Mercedes Benz to incorporate various aspects of mobility and provide luxury at a level that is yet to be experienced by users. The prime characteristics of the

car interior include four rotating lounge chairs with the ability to swing outward by 30 degrees upon opening of the doors. Apart from the car interior, the key feature of this autonomous vehicle is to enable the information exchange between the outside world, vehicles, and passengers using six display screens. The Mercedes-Benz F 015 uses visual and acoustic communication to enable communication with its surroundings. The F 015 will use laser and light emitting diodes in the front and rear end to visually communicate with other vehicles and pedestrians. This model will use 3D video cameras, lasers, infrared sensors, and radar of different ranges, such as low, high, and mid-range, to enable its autonomous operation. Although the Mercedes-Benz F 015 provides several benefits, such as lane-keeping assistance, adaptive cruise control with brake automation, and self-steering, it is still in its infancy.

TESLA AUTOPILOT

The Tesla Autopilot (accessed on: 24 Oct. 2018 <https://www.theverge.com/2018/10/20/18000884/tesla-full-self-driving-option-gone-musk-autopilot>) is envisioned to offer an advanced driving assistance system for enabling self-parking, adaptive cruise control, lane centering, and automatic lane changing functions. In 2014, the first Autopilot was introduced for the Tesla Model S, which offered only self-parking and semi-autonomous driving. Then, the Tesla Autopilot version 7.0 was released in October 2015, followed by the Autopilot 8.0 in August 2016. In the Autopilot 8.0, radar is considered a primary sensor compared with a camera. The latest version of Tesla (9.0) offers on-ramp to off-ramp features. Meanwhile, hardware 1 (2014 Autopilot) of the initial version of the Autopilot included one monochrome camera, rear view camera to enable human usage rather than automation, and 12 sonars. In hardware 2 (2016 enhanced Autopilot), forward radar, forward cameras, forward-looking side cameras, rearward looking side cameras, rearview camera, and 12 sonars are considered. In 2018, a Tesla Autopilot crashed into a stationary vehicle. The manual reveals that this accident is caused by the traffic aware cruise control, which is unable to detect all objects, especially when the speed is higher than 50 mph.

BMW VISION iNEXT

The BMW VISION iNEXT (accessed on: 21 October 2018 <https://www.bmwgroup.com/BMW-Vision-iNEXT>) is a concept of an autonomous car presented at CES 2016 to offer users fully connected, fully automatic, and highly autonomous driving services. This program is a joint project of Mobileye, Intel, and BMW and is planned to debut on roads by 2021. The driver of a BMW VISION iNEXT can freely select whether to drive or be driven. The car is envisioned to provide entertainment, interactive services, and relaxation through panoramic roof floods, seats in front of each other, and a bench seat in the rear end. The driver area consists of a steering wheel with two digital displays. The car can be operated with two modes, boost and ease, depending on the involvement level of the driver. The former is for classic, manual driving, while the latter makes it a self-driving car. The BMW VISION iNEXT is

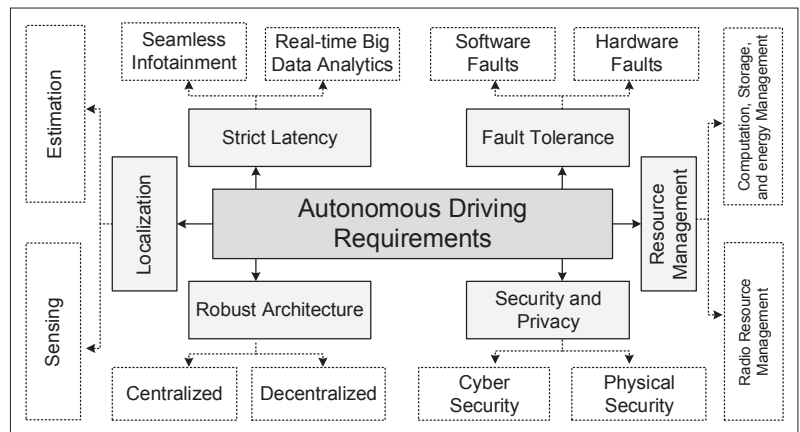


FIGURE 5. Autonomous driving requirements.

planned to be developed using multiple sensors, battery, and chip-based lidar. Moreover, the BMW VISION iNEXT aims to employ Shy technology in three different forms: intelligent beam, intelligent materials, and intelligent personal assistant.

TOYOTA LEXUS LS 600hL

Toyota has launched the Lexus LS 600hL (accessed on: 28 Oct. 2018 <https://www.theverge.com/2018/1/4/16849422/toyota-self-driving-car-platform-3-lexus-lidar>) self-driving car. The car has several features, such as an adaptive front lighting system, high-end navigation system, intuitive park assist system, blind spot monitoring, pre-collision detection system, collision avoidance, radar cruise control, and a lane-keeping assist system. This version of Toyota self-driving car demonstrates an improvement over its old version in terms of the farther visibility feature. A set of four long-range lidars developed by LUMINAR is used on the car roof to enable farther visibility. A set of cameras along with lidars are also used under a weather-proof panel. Short range lidar sensors are mounted on the rear and front bumpers of the car. These low mounted sensors consider the safety of children and detect unusual activities on roads.

OPEN RESEARCH CHALLENGES

This section discusses five imperative research challenges hindering the successful deployment of autonomous cars. The purpose is to provide guidelines with which new researchers and developers working in the domain can resolve the challenges. Table 2 summarizes the challenges along with certain guidelines.

SECURITY

Although numerous evident issues still need to be resolved before self-driving vehicles become a mainstream presence on roads, security is one of the intriguing issues among them. Autonomous cars are usually based on cyber-physical systems because it exhibits elements in physical and virtual environments, thereby making security maintenance highly challenging. Autonomous cars are vulnerable to not only traditional cyber attacks (such as GPS jamming and spoofing, millimeter-wave radar attacks, lidar sensor attacks, ultrasonic sensor attacks, camera sensor attacks, key fob cloning, telematics service attacks, map modification via an update, replay retrieval, blind range sensor, and DoS parking space

Challenge	Causes	Guidelines
Security	• Lack of sophisticated software solutions. • Lack of frequent software updates. • Involvement of cyber-physical systems.	• Quantum cryptography and blockchain enabled security algorithms to mitigate internal and external threats associated with sensors. • Defense in depth mindset. • Properly analyzing logs. • Deploying machine learning models.
Radar interference management	• High road traffic. • Lack of collision detection and avoidance algorithms. • Multiple radars operating in overlapping bands are within the same vicinity.	• Dealing with RF noise and radar that identifies and classifies obstacles.
Heterogeneous vehicular networks	• Involvement of a large number of connected cars. • Limited network capacity. • Rapid change in topology.	• Integrating different wireless access networks based on heterogeneous infrastructures.
Artificial intelligence for autonomous driving cars	• Lack of real-time data analysis solutions. • Complex datasets of heterogeneous types.	• Convergence of edge computing and AI. • Data fusion and deep learning based solutions.
Edge-assisted autonomous driving	• Scalability issues in terms of functionality, administration, and load. • A massive amount of data production.	• High-performance memory. • Deploying edge nodes at most suitable places.

TABLE 2. Open research challenges along with guidelines.

allocator), but also involve several risks from new breeds of attacks (such as ransomware and vehicle theft). Autonomous cars are usually connected through heterogeneous networks (such as electricity infrastructure, sensor networks deployed along the roadsides, and traffic control systems), which also involve security risks. In the future, robust and fool-proof defense solutions for autonomous cars must be extensively developed to make them widely adopted in smart cities.

RADAR INTERFERENCE MANAGEMENT

Lasers are usually mounted on the rooftop and body of autonomous cars for navigational purposes. The radar helps to detect reflections of radio waves from the surrounding objects. The distance between the car and the object is measured on the basis of the time needed for the reflection to manifest, which leads to the execution of appropriate action. When radar technology is utilized for thousands of connected cars, the interference problem occurs, thereby resulting in inefficient spectrum utilization. The interference problem can enable dangerous blinding of the radars in dense urban communities, thereby hindering the widespread deployment of autonomous cars. A solution, called RadarMAC, has been proposed in [14] to resolve the radar interference problem. However, the proposed solution is in its infancy, and thus additional solutions need to be proposed for handling the radar interference problem.

HETEROGENEOUS VEHICULAR NETWORKS

Autonomous cars are highly mobile, thereby causing rapid changes in topology and consequently hindering adequate services using a single wireless access network, such as DSRC and LTE. The DSRC networks were mainly designed to support short-range communications without considering the pervasive communication infrastructure. Real-time information is one of the prime requirements of autonomous cars, which cannot be fulfilled using LTE networks due to the involvement of a considerable number of cars. Meanwhile, auto-

nomous cars typically generate a massive amount of data, which is beyond the managing capabilities of traditional vehicular networks. One possible solution of the above-mentioned challenges is to integrate different wireless access networks by forming heterogeneous vehicular networks, thereby satisfying certain communication challenges of autonomous cars [3]. However, integration of the multiple vehicular networks engenders new challenges, which must be addressed.

ARTIFICIAL INTELLIGENCE FOR AUTONOMOUS DRIVING CARS

Although AI-based autonomous cars can never be exactly analogous to the human brain, modern AI algorithms can help in making roads safe and efficient in terms of traveling. For example, deep learning-based solutions can help identify unforeseen obstacles on the roads ahead, and reinforcement learning can enable cars to control motion and help in the decision-making process [15]. However, development of such solutions that can perform real-time analysis of the massive amount of diverse data collected from autonomous cars becomes taxing. Strict latency requirements can be compensated by bringing analytic capabilities near autonomous cars through the edge computing concept [5]. Subsequently, data fusion-based solutions along with modern AI algorithms can help in accurately analyzing the data, thereby leading to enhanced decision-making capabilities. Specifically, the convergence of edge computing and AI can be an ideal solution for obtaining real-time insights into the vehicles' data. However, this convergence engenders new issues that must be resolved in the future.

EDGE-ASSISTED AUTONOMOUS DRIVING CARS

Cloud computing has empowered autonomous cars to efficiently store and process their data. However, the limited bandwidth of wireless networks often leads to performance degradation of cloud computing for autonomous cars because they efficiently transmit a massive amount of data. The network performance bottleneck can be alleviated through the edge computing concept,

which envisions to bring computation close to the vehicles, thereby resulting in meeting various indispensable requirements, such as reliability, timing, power, and energy consumption, of autonomous vehicles [5]. However, bringing services near the vehicles' network where connectivity of the cars and their data is increasing at a tremendous rate often becomes highly crucial due to scalability issues in terms of functionality, administration, and load. Moreover, the connectivity among a large number of devices results in a flood of data production that can hinder the edge node to perform analytics on such a large-scale data by meeting strict latency requirements of autonomous cars. An adequate consideration must be given to resolve the edge-related issues for enabling successful deployment of autonomous cars.

CONCLUSION

In this article, we initially investigated, highlighted, and reported premier research advances made in autonomous driving by devising a taxonomy. Then, we enumerated and discussed six indispensable requirements for the successful deployment of self-driving cars. We discovered and presented recent synergies and prominent case studies on autonomous cars. Furthermore, we unearthed and discussed future research challenges. We conclude that although autonomous cars are assisting people by providing reliable and safe transportation services, their deployment engenders imperative challenges that must be addressed in the future.

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BIOGRAPHY

IBRAR YAQOOB (S'16, M'18, SM'19) is a research professor with the Department of Computer Science and Engineering, Kyung Hee University, South Korea, where he completed his post-doctoral fellowship. He received the Ph.D. (computer science) degree from the University of Malaya, Malaysia. His numerous research articles are very famous and among the most downloaded in top journals. He is serving as a guest/associate editor in various journals. He has been involved in a number of conferences and workshops in various capacities. His research interests include big data, edge computing, mobile cloud computing, Internet of Things, and computer networks.

LATIF U. KHAN is pursuing his Ph.D. degree in computer engineering at Kyung Hee University (KHU), South Korea. He is working as a leading researcher in the Intelligent Networking Laboratory under a project jointly funded by the prestigious Brain Korea 21st Century Plus and Ministry of Science and ICT, South Korea. He received his MS (electrical engineering) degree with distinction from the University of Engineering and Technology (UET), Peshawar, Pakistan in 2017. His research interests include analytical techniques of optimization and game theory to edge computing and end-to-end network slicing.

S. M. AHSAN KAZMI received his Ph.D. degree in computer science and engineering from Kyung Hee University, South Korea in 2017. Since 2018, he has been with the Institute of Information Systems (IIS), Innopolis University, Innopolis, Tatarstan, Russia, where he is currently an assistant professor. His research interests include applying analytical techniques of optimization and game theory to radio resource management for future cellular networks. He received the best KHU Thesis Award in engineering in 2017 and several best paper awards from prestigious conferences.

MUHAMMAD IMRAN is currently working at King Saud University and is a visiting scientist with Iowa State University. His research interests include MANET, WSNs, WBANs, M2M/IoT, SDN, security and privacy. He has published a number of research papers in refereed international conferences and journals. He serves as a co-editor in chief for EAI transactions and an associate/guest editor for *IEEE Access*, *IEEE Communications Magazine*, *Computer Networks*, *Future Generation Computer Systems*, *Sensors*, *IJDSN*, *JIT*, *WCMC*, *AHSWN*, *IET WSS*, *IJAACS* and *IJITEE*.

NADRA GUIZANI is currently working at Gonzaga University. She served as a teaching assistant in the Department of Electrical and Computer Engineering, Purdue University. Her research work is on data analytics and prediction and access control of disease spread data on dynamic network topologies. Research interests include machine learning, mobile networking, large data analysis, and prediction techniques.

CHOONG SEON HONG [S'95, M'97, SM'11] is working as a professor with the Department of Computer Science and Engineering, Kyung Hee University. His research interests include future Internet, ad hoc networks, network management, and network security. He is a member of ACM, IEICE, IPSJ, KIISE, KICS, KIPS, and OSIA. He has served as the General Chair, a TPC Chair/Member, or an Organizing Committee Member for international conferences such as NOMS, IM, APNOMS, E2EMON, CCNC, ADSN, ICPP, DIM, WISA, BcN, TINA, SAINT, and ICOIN. In addition, he is currently an associate editor of the *IEEE Transactions on Network and Service Management*, the *International Journal of Network Management*, and the *Journal of Communications and Networks* and an associate technical editor of *IEEE Communications Magazine*.